

Business Case: Walmart - Confidence Interval and CLT

1.Introduction

About Walmart ?

Walmart is an American multinational retail corporation that operates a chain of supercenters, discount departmental stores, and grocery stores from the United States. Walmart has more than 100 million customers worldwide.

Objective:

The objective of this project is to conduct a comprehensive analysis of customer purchase behavior, with a specific focus on purchase amounts, in relation to customer gender during the Black Friday sales event at Walmart Inc. This study aims to provide valuable insights that can assist the management team at Walmart Inc. in making data-driven decisions.

2.Exploratory Data Analysis

```
#importing libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import t
import warnings
warnings.filterwarnings('ignore')
import copy
```

```
df=pd.read_csv("walmart_data.txt")
```

```
df.head()
```

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
0	1000001	P00069042	F	0-17	10	A	2	0	3	8370
1	1000001	P00248942	F	0-17	10	A	2	0	1	15200
2	1000001	P00087842	F	0-17	10	A	2	0	12	1422

```
df.shape
```

```
(550068, 10)
```

```
df.isnull().sum() #No missing values in dataset
```

```
User_ID          0
Product_ID       0
Gender           0
Age              0
Occupation       0
City_Category    0
Stay_In_Current_City_Years  0
Marital_Status   0
Product_Category 0
Purchase         0
dtype: int64
```

```
df.nunique()
```

```
User_ID          5891
Product_ID       3631
Gender           2
Age              7
Occupation       21
City_Category    3
Stay_In_Current_City_Years  5
Marital_Status   2
Product_Category 20
Purchase        18105
dtype: int64
```

Insights

- There are total **550068** datapoints out of which only **5891** User_ID found means same users bought multiple product or do multiple transactions.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   User_ID             550068 non-null  int64
 1   Product_ID          550068 non-null  object
 2   Gender              550068 non-null  object
 3   Age                 550068 non-null  object
 4   Occupation          550068 non-null  int64
 5   City_Category       550068 non-null  object
 6   Stay_In_Current_City_Years  550068 non-null  object
 7   Marital_Status      550068 non-null  int64
 8   Product_Category    550068 non-null  int64
 9   Purchase            550068 non-null  int64
dtypes: int64(5), object(5)
memory usage: 42.0+ MB
```

No. of categorical feature - 5

No. of continuous (numerical) feature - 5

```
cat_cols = ["User_ID", "Product_ID", "Gender", "Age", "City_Category", "Marital_Status"]
```

```
for i in cat_cols:
    df[i] = df[i].astype("category")
```

```
df.dtypes
```

```
User_ID           category
Product_ID        category
Gender            category
Age              category
Occupation         int64
City_Category     category
Stay_In_Current_City_Years  object
Marital_Status    category
Product_Category  int64
Purchase          int64
dtype: object
```

```
df.describe()
```

	Occupation	Product_Category	Purchase	
count	550068.000000	550068.000000	550068.000000	
mean	8.076707	5.404270	9263.968713	
std	6.522660	3.936211	5023.065394	
min	0.000000	1.000000	12.000000	
25%	2.000000	1.000000	5823.000000	
50%	7.000000	5.000000	8047.000000	
75%	14.000000	8.000000	12054.000000	
max	20.000000	20.000000	23961.000000	

- As it is quite obvious that there is significant difference in the mean and std for Purchase column which might be containing outliers.

```
df.describe(include = ['object','category'])
```

	User_ID	Product_ID	Gender	Age	City_Category	Stay_In_Current_City_Years	Marital_Status	
count	550068	550068	550068	550068	550068	550068	550068	
unique	5891	3631	2	7	3	5	2	
top	1001680	P00265242	M	26-35	B	1	0	
freq	1026	1880	414259	219587	231173	193821	324731	

- Customer (1001680) has purchased more than others
- Product (P00265242) is most bought item
- Most of the customers are Male
- Most of customers lies in [26-35] Age bracket
- Majority of the customers are Unmarried

```
df.duplicated().sum()

0
```

Non-Graphical Analysis

Value counts and Unique Attributes

```
df.groupby(["Gender"])["User_ID"].nunique() / df["User_ID"].nunique() * 100
```

Gender
F 28.280428
M 71.719572
Name: User_ID, dtype: float64

```
df.groupby(["Gender"])["Purchase"].describe()
```

	count	mean	std	min	25%	50%	75%	max	
Gender									
F	135809.0	8734.565765	4767.233289	12.0	5433.0	7914.0	11400.0	23959.0	
M	414259.0	9437.526040	5092.186210	12.0	5863.0	8098.0	12454.0	23961.0	

- There are only 28% female customers. While male customers constitute around 72% in our dataset

```
df.groupby(["Age"])["User_ID"].nunique() / df["User_ID"].nunique() * 100
```

```
Age
0-17      3.700560
18-25     18.146325
26-35     34.849771
36-45     19.809879
46-50      9.013750
51-55      8.164997
55+        6.314717
Name: User_ID, dtype: float64
```

```
df.groupby(["Age"])[ "Purchase" ].describe()
```

	count	mean	std	min	25%	50%	75%	max	
Age									
0-17	15102.0	8933.464640	5111.114046	12.0	5328.0	7986.0	11874.0	23955.0	
18-25	99660.0	9169.663606	5034.321997	12.0	5415.0	8027.0	12028.0	23958.0	
26-35	219587.0	9252.690633	5010.527303	12.0	5475.0	8030.0	12047.0	23961.0	
36-45	110013.0	9331.350695	5022.923879	12.0	5876.0	8061.0	12107.0	23960.0	
46-50	45701.0	9208.625697	4967.216367	12.0	5888.0	8036.0	11997.0	23960.0	
51-55	38501.0	9534.808031	5087.368080	12.0	6017.0	8130.0	12462.0	23960.0	
55+	21504.0	9336.280459	5011.493996	12.0	6018.0	8105.5	11932.0	23960.0	

- Around 35% customers are in [26-35] age bracket. On other hand customers in age bracket [0-17], [51-55] & [55+] showing less purchasing behaviour.

```
df.groupby(["Stay_In_Current_City_Years"])[ "User_ID" ].nunique()
```

```
Stay_In_Current_City_Years
0      772
1     2086
2     1145
3      979
4+      909
Name: User_ID, dtype: int64
```

- Customers staying for 1-2 years in same city purchased more than others

```
df.groupby(["Marital_Status"])[ "User_ID" ].nunique() / df["User_ID"].nunique() * 100
```

```
Marital_Status
0     58.003735
1     41.996265
Name: User_ID, dtype: float64
```

- 58:42 is the ratio between married and unmarried customers

```
df.groupby(["City_Category"])[ "User_ID"].nunique()
```

City_Category
A 1045
B 1707
C 3139
Name: User_ID, dtype: int64

- Most of customers belongs to C city_category

```
df.groupby(['City_Category'])[ 'Purchase'].describe()
```

	count	mean	std	min	25%	50%	75%	max		
City_Category										
A	147720.0	8911.939216	4892.115238	12.0	5403.0	7931.0	11786.0	23961.0		
B	231173.0	9151.300563	4955.496566	12.0	5460.0	8005.0	11986.0	23960.0		
C	171175.0	9719.920993	5189.465121	12.0	6031.5	8585.0	13197.0	23961.0		

- Most of the customers belongs to City C But customers from city_category B tend to purchase more suggest the fact they visit store multiple times during that period

```
categorical_cols = ['Gender', 'Age', 'Occupation', 'City_Category', 'Stay_In_Current_City_Years', 'Marital_Status', 'Product_Category']  
df[categorical_cols].melt().groupby(['variable', 'value'])['value'].count()/len(df)
```

		value	
variable	value		
Age	0-17	0.027455	
	18-25	0.181178	
	26-35	0.399200	
	36-45	0.199999	
	46-50	0.083082	
	51-55	0.069993	
	55+	0.039093	
City_Category	A	0.268549	
	B	0.420263	
	C	0.311189	
Gender	F	0.246895	
	M	0.753105	
Marital_Status	0	0.590347	
	1	0.409653	
Occupation	0	0.126599	
	1	0.086218	
	2	0.048336	
	3	0.032087	
	4	0.131453	
	5	0.022137	
	6	0.037005	
	7	0.107501	
	8	0.002811	
	9	0.011437	
	10	0.023506	
	11	0.021063	
	12	0.056682	
	13	0.014049	
	14	0.049647	
	15	0.022115	
	16	0.046123	

Product_Category	17	0.072796
	18	0.012039
	19	0.015382
	20	0.061014
	1	0.255201
	2	0.043384
	3	0.036746
	4	0.021366
	5	0.274390
	6	0.037206
	7	0.006765
	8	0.207111
	9	0.000745
	10	0.009317
	11	0.044153
	12	0.007175
	13	0.010088
	14	0.002769
	15	0.011435
	16	0.017867
	17	0.001051
Stay_In_Current_City_Years	18	0.005681
	19	0.002914
	20	0.004636
	0	0.135252
	1	0.352358
	2	0.185137
	3	0.173224
	4+	0.154028

Observations

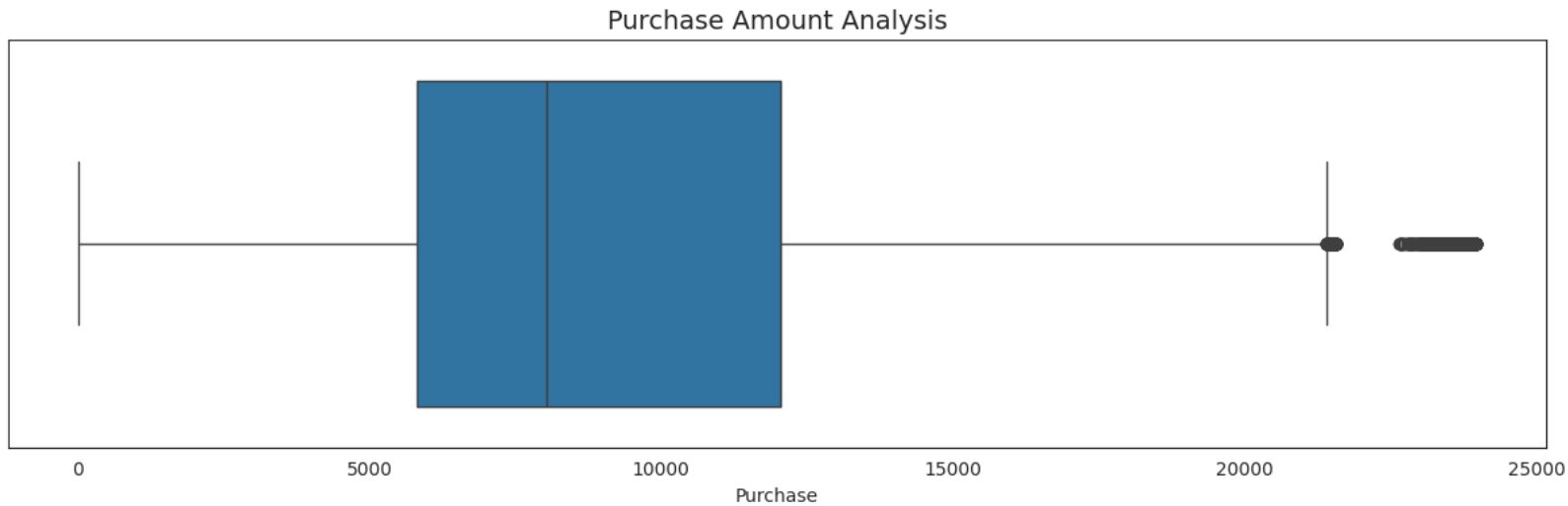
- ~ 80% of the users are between the age 18-50 (40%: 26-35, 18%: 18-25, 20%: 36-45)
- 75% of the users are **Male** and 25% are **Female**

- 60% Single, 40% Married
- 35% Staying in the city from 1 year, 18% from 2 years, 17% from 3 years
- Total of 20 product categories are there
- There are 20 different types of occupations in the city

✓ 3.Graphical Analysis

✓ Univariate Countinous Variable Analysis

```
fig = plt.figure(figsize=(15,4))
sns.set_style("white")
sns.boxplot(data = df, x = "Purchase", orient = "h")
plt.title('Purchase Amount Analysis', fontsize = '14')
plt.show()
```



Insights

- There are outliers in purchase amount.
- While observing the distribution of purchase amount from density plot. It is quite obvious that the distribution is right skewed means majority of data concentrated on left side.
- Majority of customer purchase within 5,000 - 20,000 range.

✓ Detecting Outliers

```
print('Mean of Purchase Amount    = ', df["Purchase"].mean())
print('Median of Purchase Amount = ', df["Purchase"].median()) #Lot of Difference in Mean and Median tells there are outliers
```

```
Mean of Purchase Amount    =  9263.968712959126
Median of Purchase Amount =  8047.0
```

✓ Handling Outliers

```
q1 = df["Purchase"].quantile(0.25)
q3 = df["Purchase"].quantile(0.75)
```

```
IQR = q3 - q1
print("1. First Quartile (q1) = ", q1)
print("2. Third Quartile (q2) = ", q3)
print("3. IQR                = ", IQR)
```

```
1. First Quartile (q1) =  5823.0
2. Third Quartile (q2) = 12054.0
3. IQR                =  6231.0
```

```
df_new = df.copy()
```

```
df_new["Purchase"] = np.clip(df_new["Purchase"], q1-1.5*IQR, q3 + 1.5*IQR)
df_new.shape
```

```
(550068, 10)
```

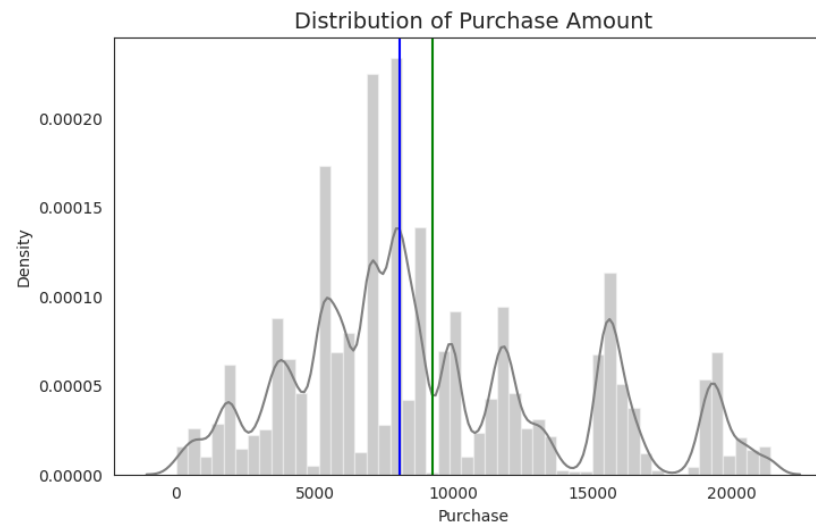
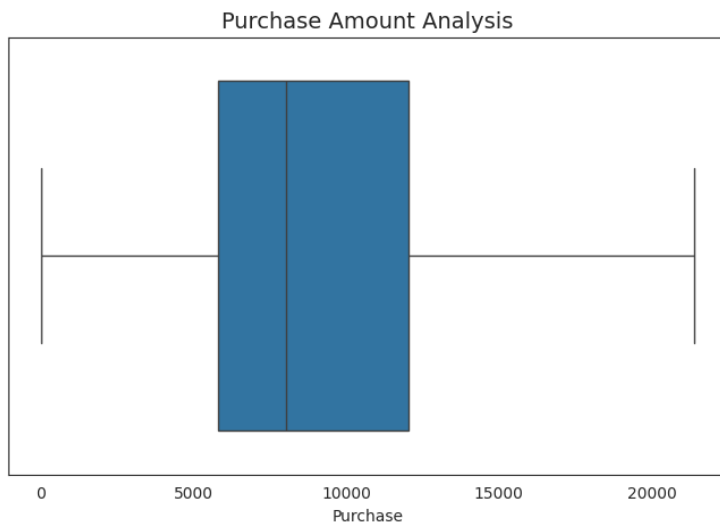
```
fig = plt.figure(figsize=(18,5))
sns.set_style("white")
```

```
plt.subplot(1, 2, 1)
sns.boxplot(data = df_new, x = "Purchase", orient = "h")
```

```
plt.title('Purchase Amount Analysis', fontsize = '14')
```

```
plt.subplot(1, 2, 2)
sns.distplot(a = df_new["Purchase"], color = 'grey')
plt.title("Distribution of Purchase Amount", fontsize = '14')
plt.axvline(df_new["Purchase"].mean(),color="g")
plt.axvline(df_new["Purchase"].median(),color="b")
```

```
plt.show()
```



Outliers are Handled using IQR and updating the dataset values

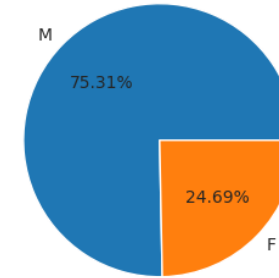
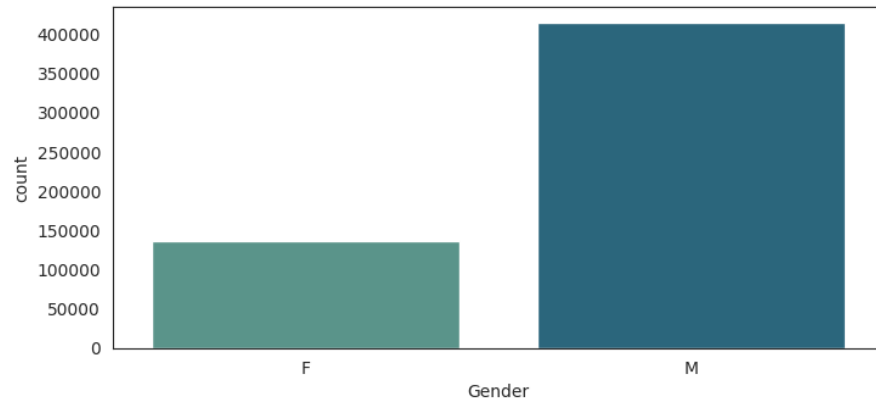
✦ Univariate Categorical Variable Analysis

```
fig = plt.figure(figsize=(18,8))
sns.set_style(style='white')
ax1 = plt.subplot2grid((2,2),(0,0))
sns.countplot(data=df, x="Gender", palette="crest")

#first row sec column
ax1 = plt.subplot2grid((2,2), (0, 1))
plt.pie(df_new["Gender"].value_counts(),
        labels = df["Gender"].value_counts().index,
        autopct = '%1.2f%%')
plt.suptitle('Distribution of Gender', fontsize = 20)

plt.show()
```

Distribution of Gender



Insights

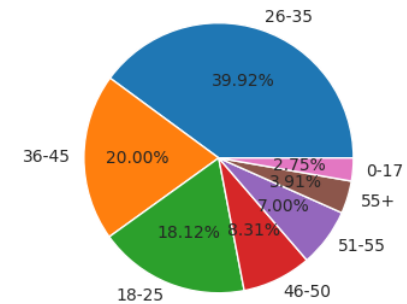
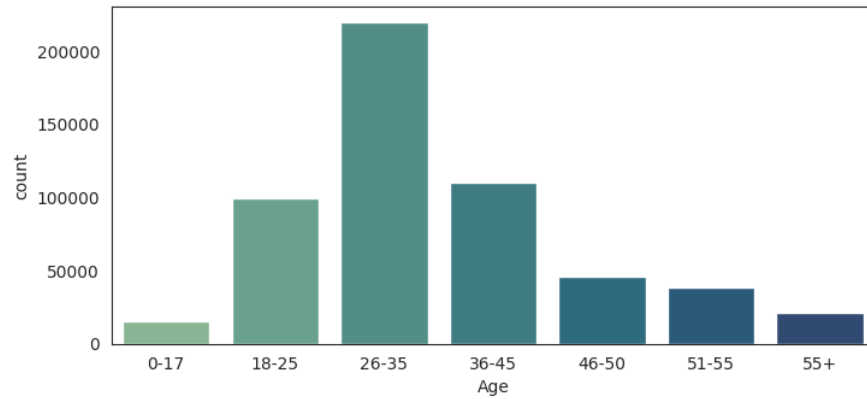
- Males clearly purchase more than females. 75% of men and only 25% of women purchase products.

```
fig = plt.figure(figsize=(18,8))
sns.set_style(style='white')
ax1 = plt.subplot2grid((2,2),(0,0))
sns.countplot(data=df, x="Age", palette="crest")

#first row sec column
ax1 = plt.subplot2grid((2,2), (0, 1))
plt.pie(df_new["Age"].value_counts(),
        labels = df["Age"].value_counts().index,
        autopct = '%1.2f%%')
plt.suptitle('Distribution of Age', fontsize = 20)

plt.show()
```

Distribution of Age



Insights

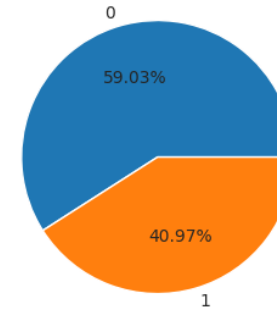
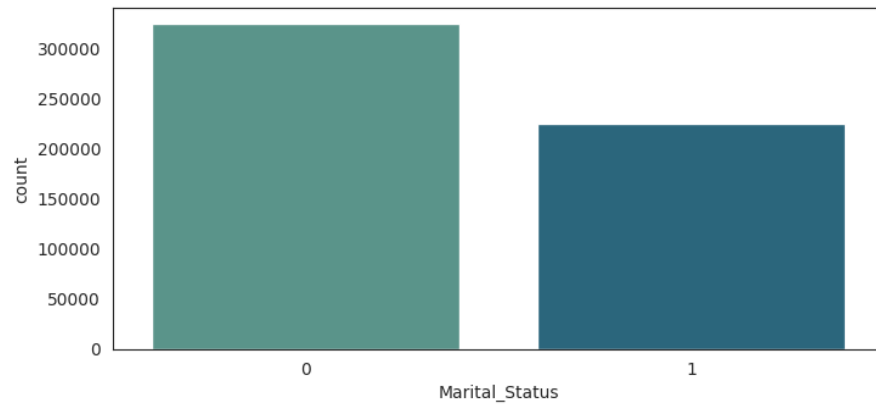
- 60% of purchases are made by people between the ages of 26 and 45

```
fig = plt.figure(figsize=(18,8))
sns.set_style(style='white')
ax1 = plt.subplot2grid((2,2),(0,0))
sns.countplot(data=df, x="Marital_Status", palette="crest")

ax1 = plt.subplot2grid((2,2), (0, 1))
plt.pie(df_new["Marital_Status"].value_counts(),
        labels = df["Marital_Status"].value_counts().index,
        autopct = '%1.2f%%')
plt.suptitle('Distribution of Marital_Status', fontsize = 20)

plt.show()
```

Distribution of Marital_Status



Insights

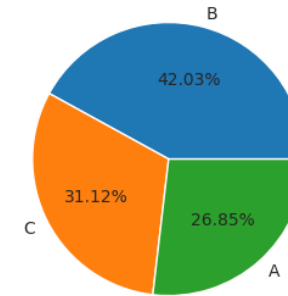
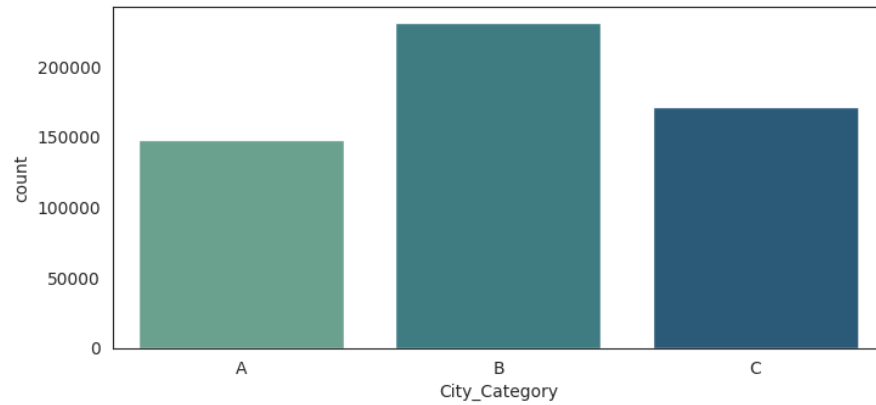
- Most of customers are unmarried.

```
fig = plt.figure(figsize=(18,8))
sns.set_style(style='white')
ax1 = plt.subplot2grid((2,2),(0,0))
sns.countplot(data=df, x="City_Category", palette="crest")

#first row sec column
ax1 = plt.subplot2grid((2,2), (0, 1))
plt.pie(df_new["City_Category"].value_counts(),
        labels = df["City_Category"].value_counts().index,
        autopct = '%1.2f%%')
plt.suptitle('Distribution of City_Category', fontsize = 20)

plt.show()
```

Distribution of City_Category



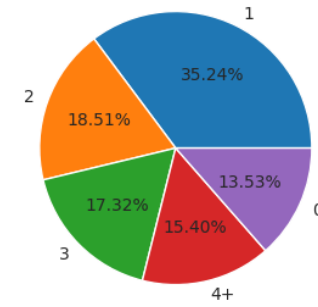
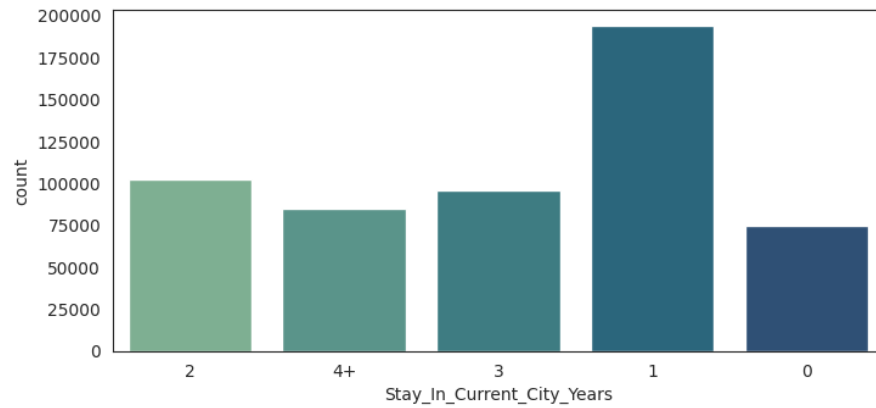
Insights

- City Category B accounts for 42%, City Category C 31%, and City Category A represents 27% of all customer purchases.

```
fig = plt.figure(figsize=(18,8))
sns.set_style(style='white')
ax1 = plt.subplot2grid((2,2),(0,0))
sns.countplot(data=df, x="Stay_In_Current_City_Years", palette="crest")

#first row sec column
ax1 = plt.subplot2grid((2,2), (0, 1))
plt.pie(df_new["Stay_In_Current_City_Years"].value_counts(),
        labels = df["Stay_In_Current_City_Years"].value_counts().index,
        autopct = '%1.2f%%')
plt.suptitle('Distribution of Stay_In_Current_City_Years', fontsize = 20)
plt.show()
```

Distribution of Stay_In_Current_City_Years

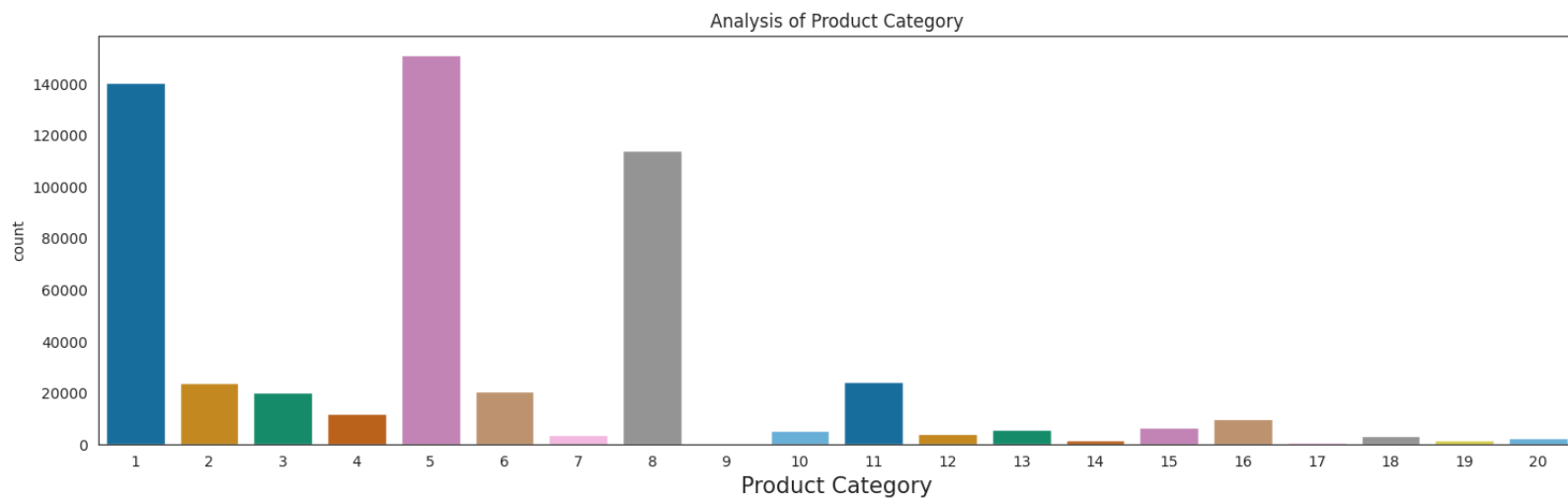


Insights

- Most of customer stayed around 1 years in the current city

```
fig = plt.figure(figsize = (18,5))

sns.countplot(data = df, x = "Product_Category", palette="colorblind")
plt.xlabel("Product Category", fontsize = 15)
plt.title("Analysis of Product Category")
plt.show()
```



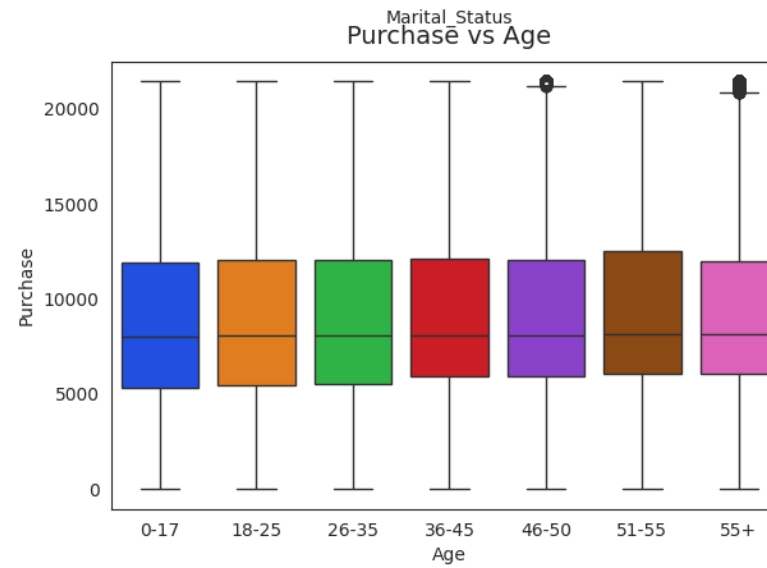
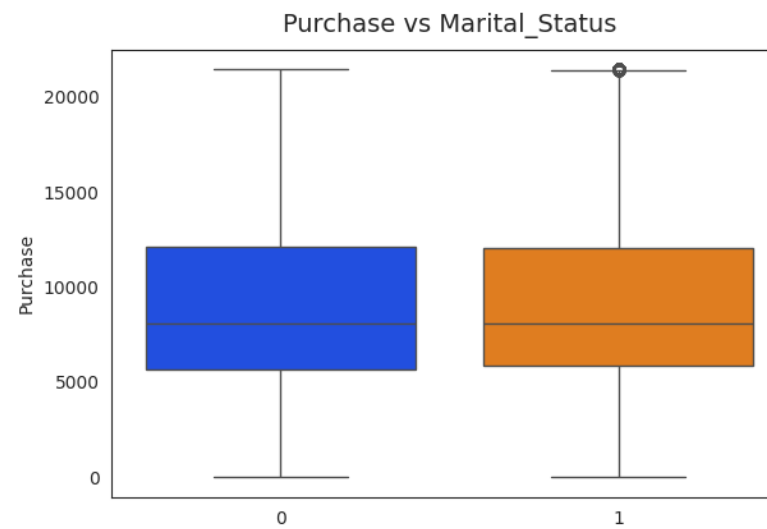
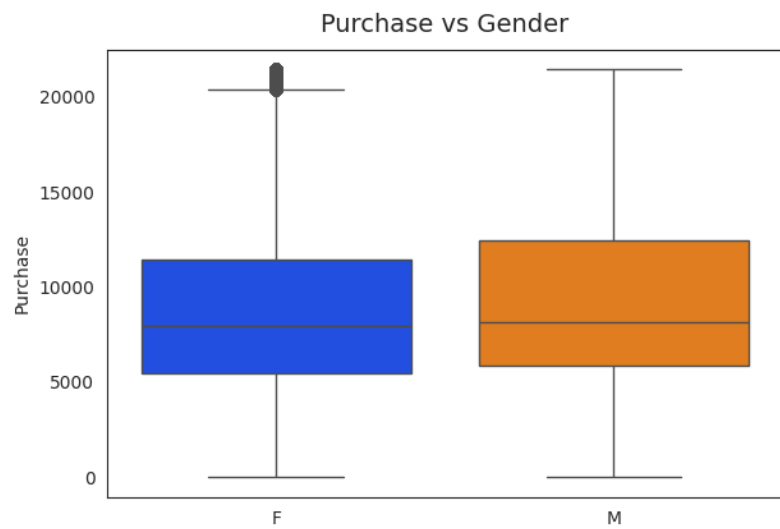
Insights

- Product categories 1, 5 and 8 most bought products.

✓ Bivariate Analysis

```
cat_col = ["Gender", "Marital_Status", "City_Category", "Age"]

fig, axs = plt.subplots(nrows=2, ncols = 2, figsize=(15,10))
k = 0
sns.set_style("dark")
for i in range(2):
    for j in range(2):
        sns.boxplot(data=df_new, x=cat_col[k], y="Purchase", palette='bright', ax=axs[i, j])
        axs[i, j].set_title("Purchase vs " + cat_col[k], pad = 10, fontsize = 14)
        k += 1
plt.show()
```



Insights

- Customers purchasing behaviour are almost stable at equilibrium irrespective on Gender, Marital_status, City_Category & Age
- Median purchase from each features are almost same

↳ Lets draw analysis for Gender

```
fig = plt.figure(figsize=(18, 10))

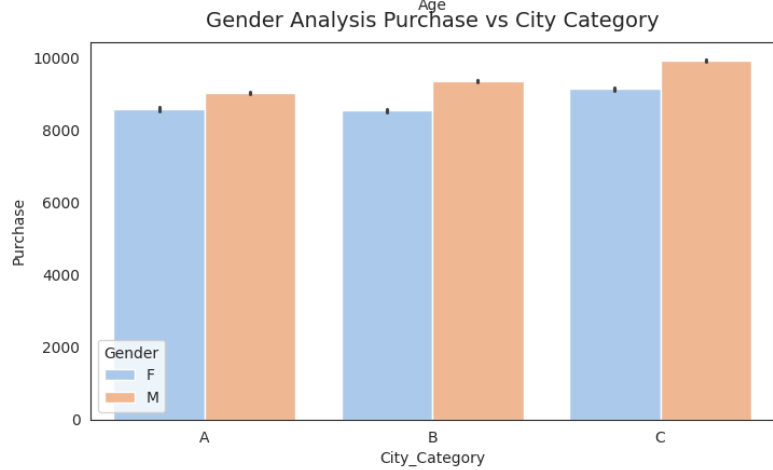
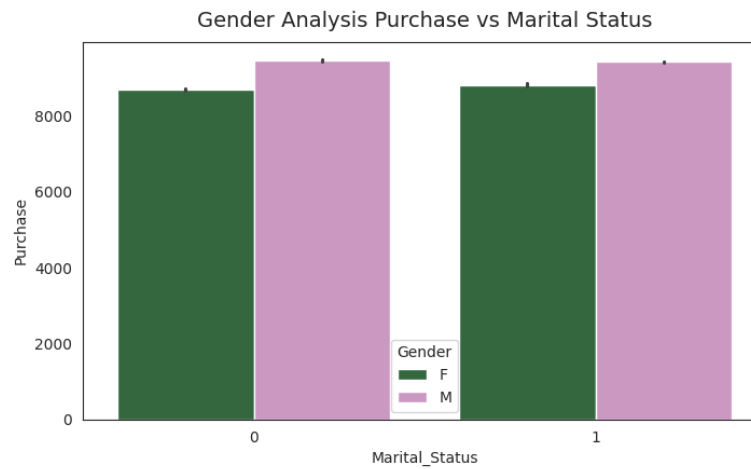
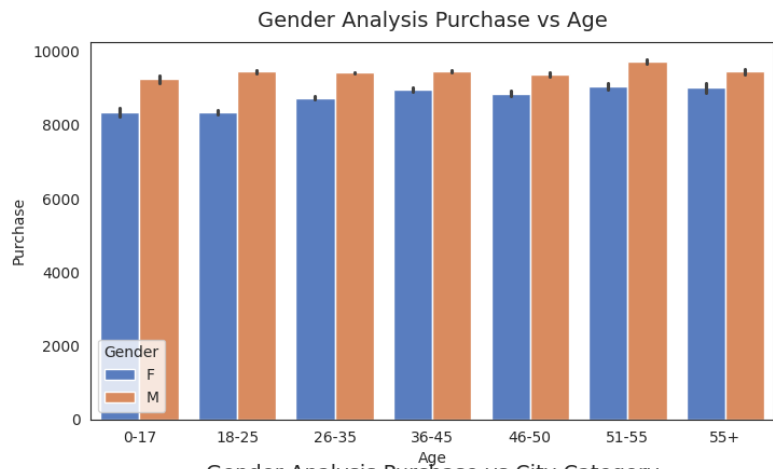
sns.set_style("white")

ax1 = plt.subplot2grid((2, 2), (0, 0))
sns.barplot(data=df, x = "Age", y = "Purchase", ax=ax1, hue = "Gender", palette="muted")
plt.title("Gender Analysis Purchase vs Age", pad = 10, fontsize = 14)

ax1 = plt.subplot2grid((2, 2), (0, 1))
sns.barplot(data=df, x = "Marital_Status", y = "Purchase", ax=ax1, hue = "Gender", palette="cubehelix")
plt.title("Gender Analysis Purchase vs Marital Status", pad = 10, fontsize = 14)

ax1 = plt.subplot2grid((2, 2), (1, 0))
sns.barplot(data=df, x = "City_Category", y = "Purchase", ax=ax1, hue = "Gender", palette="pastel")
plt.title("Gender Analysis Purchase vs City Category", pad = 10, fontsize = 14)

plt.show()
```



Insights

- Purchases are high in city category C
- Purchase is the somehow same for all age groups
- Female Customers form City category C is more than other city category
- Purchases are same for all Marital Status

```
fig = plt.figure(figsize=(18, 15))

sns.set_style("white")

ax1 = plt.subplot2grid((2, 2), (0, 0))
sns.countplot(data=df, x = "Age", ax=ax1, hue = "Gender", palette="crest")
plt.title("Age Churn Status", pad = 10, fontsize = 14)

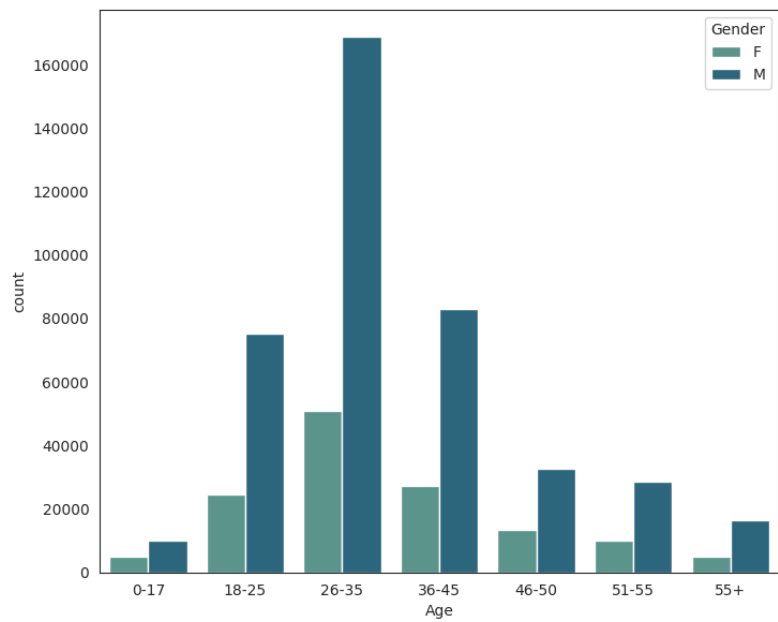
ax1 = plt.subplot2grid((2, 2), (0, 1))
sns.countplot(data=df, x = "Marital_Status", ax=ax1, hue = "Gender", palette="crest")
plt.title("Marital Churn Status", pad = 10, fontsize = 14)

ax1 = plt.subplot2grid((2, 2), (1, 0))
sns.countplot(data=df, x = "City_Category", ax=ax1, hue = "Gender", palette="crest")
plt.title("City Category Churn Status", pad = 10, fontsize = 14)

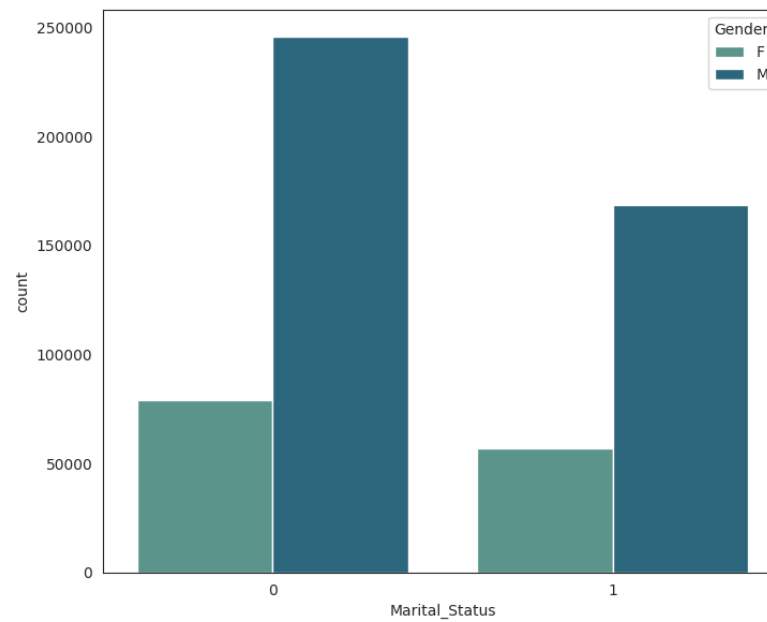
ax1 = plt.subplot2grid((2, 2), (1, 1))
sns.countplot(data=df, x = "Stay_In_Current_City_Years", ax=ax1, hue = "Gender", palette="crest")
plt.title("Stay_In_Current_City_Years Churn Status", pad = 10, fontsize = 14)

plt.show()
```

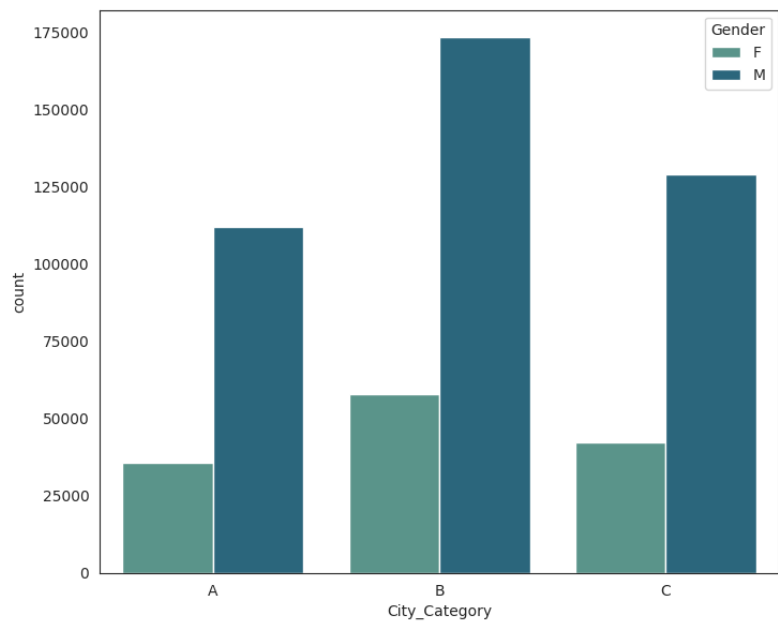
Age Chrun Status



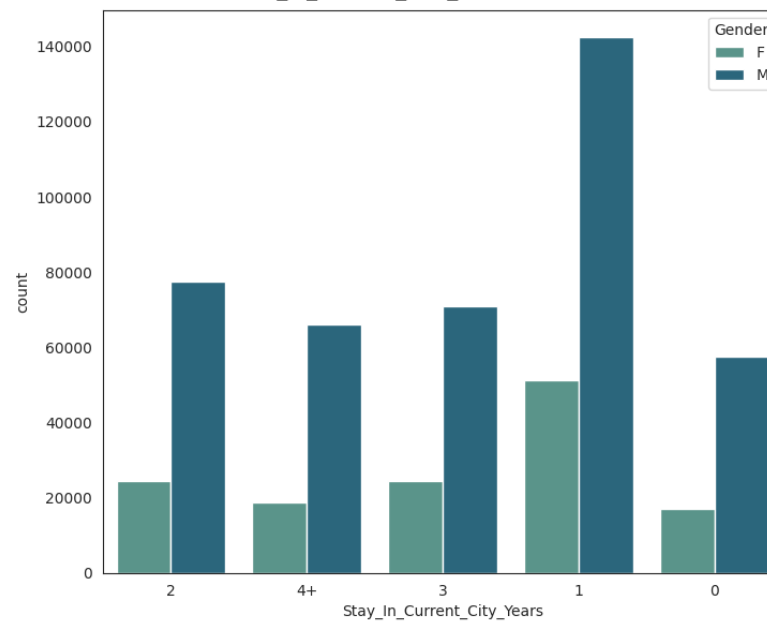
Marital Chrun Status



City Category Churn Status



Stay_In_Current_City_Years Churn Status



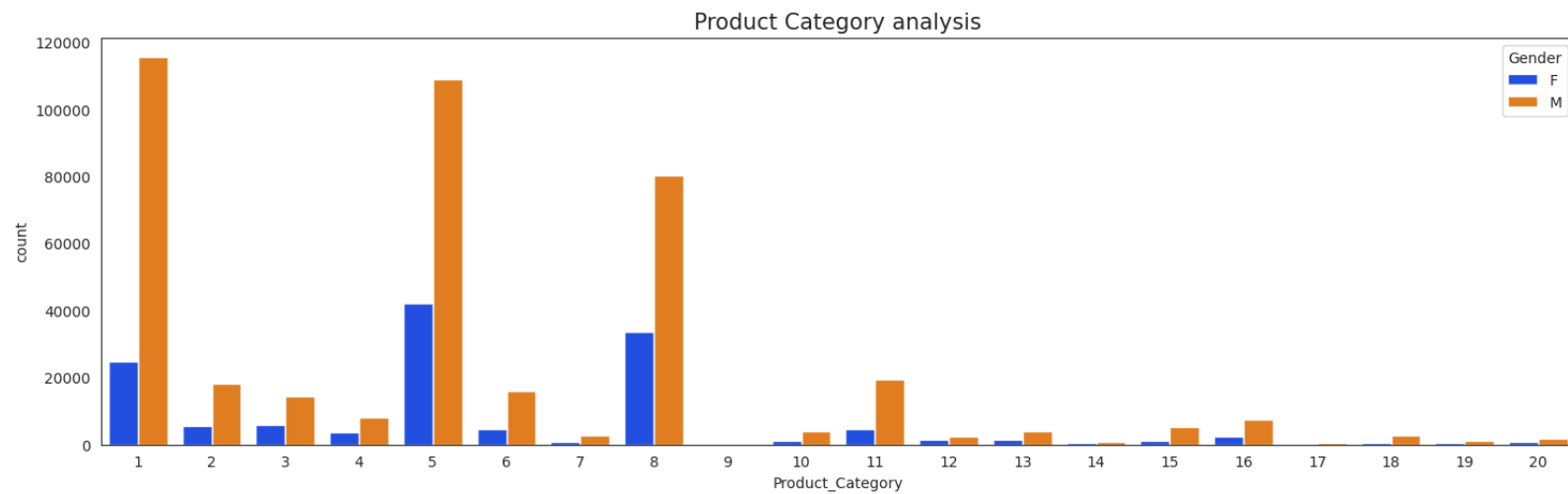
Insights

- Most of female customers are from city category C
- Most of female customer stay 1 year in the current city than others
- Most of the customers are in range 18-45 age bracket
- There is huge difference in the male and female customer for age bracket [26-35]
- There is significant difference of male customers. Most of the male customers are single. While female customers marital status not more significant as there is not so much difference.

```
fig = plt.figure(figsize=(18, 5))

plt.title("Product Category analysis", fontsize = 15)
sns.countplot(data=df, x = "Product_Category", hue = "Gender", palette="bright")

plt.show()
```



Insights

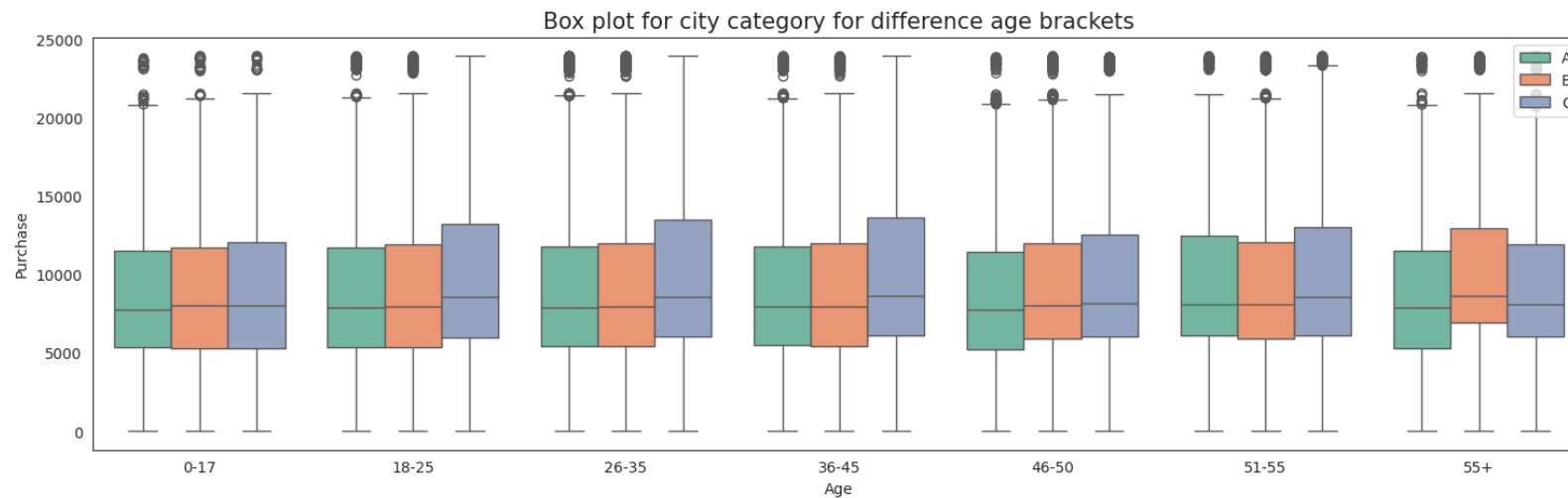
- Product Category 5 and 8 are demanding among female customers
- Proudct Category 1 and 5 are most demanding among male customers

✓ Lets draw analysis for City Category Among Different Age Brackets

```
fig = plt.figure(figsize=(18, 5))

plt.title("Box plot for city category for difference age brackets", fontsize = 15)
sns.boxplot(data=df, x = "Age", y = "Purchase", hue = "City_Category", palette="Set2")

plt.legend(loc = "upper right")
plt.show()
```



Insights

- Product Category B is popular among age group 55+
- Product Category A & B are quite same in demand among other age groups except 55+
- While Product Category C is in demand for age group 18-45 and slightly in 51-55 age group

Correlation Analysis

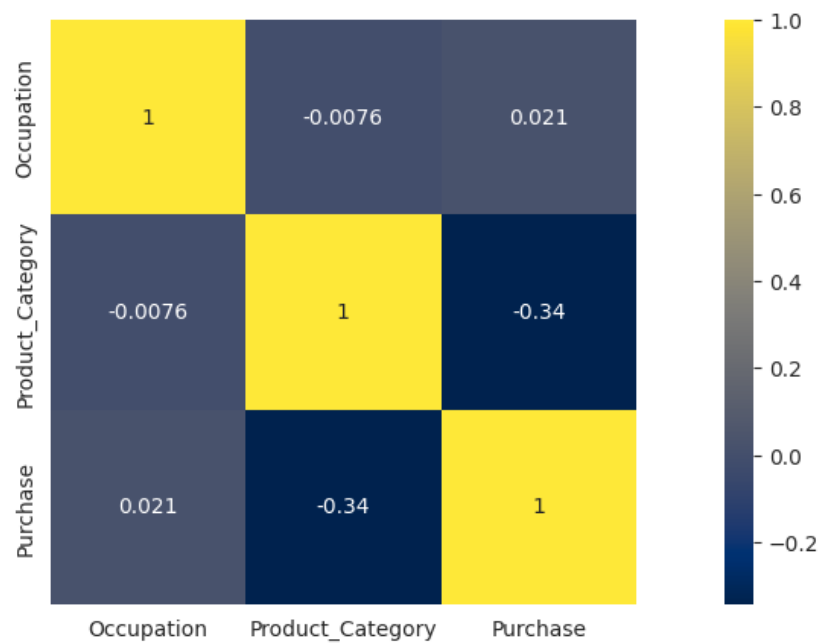
```
df_new.corr()
```

	Occupation	Product_Category	Purchase
Occupation	1.000000	-0.007618	0.020853
Product_Category	-0.007618	1.000000	-0.347413
Purchase	0.020853	-0.347413	1.000000

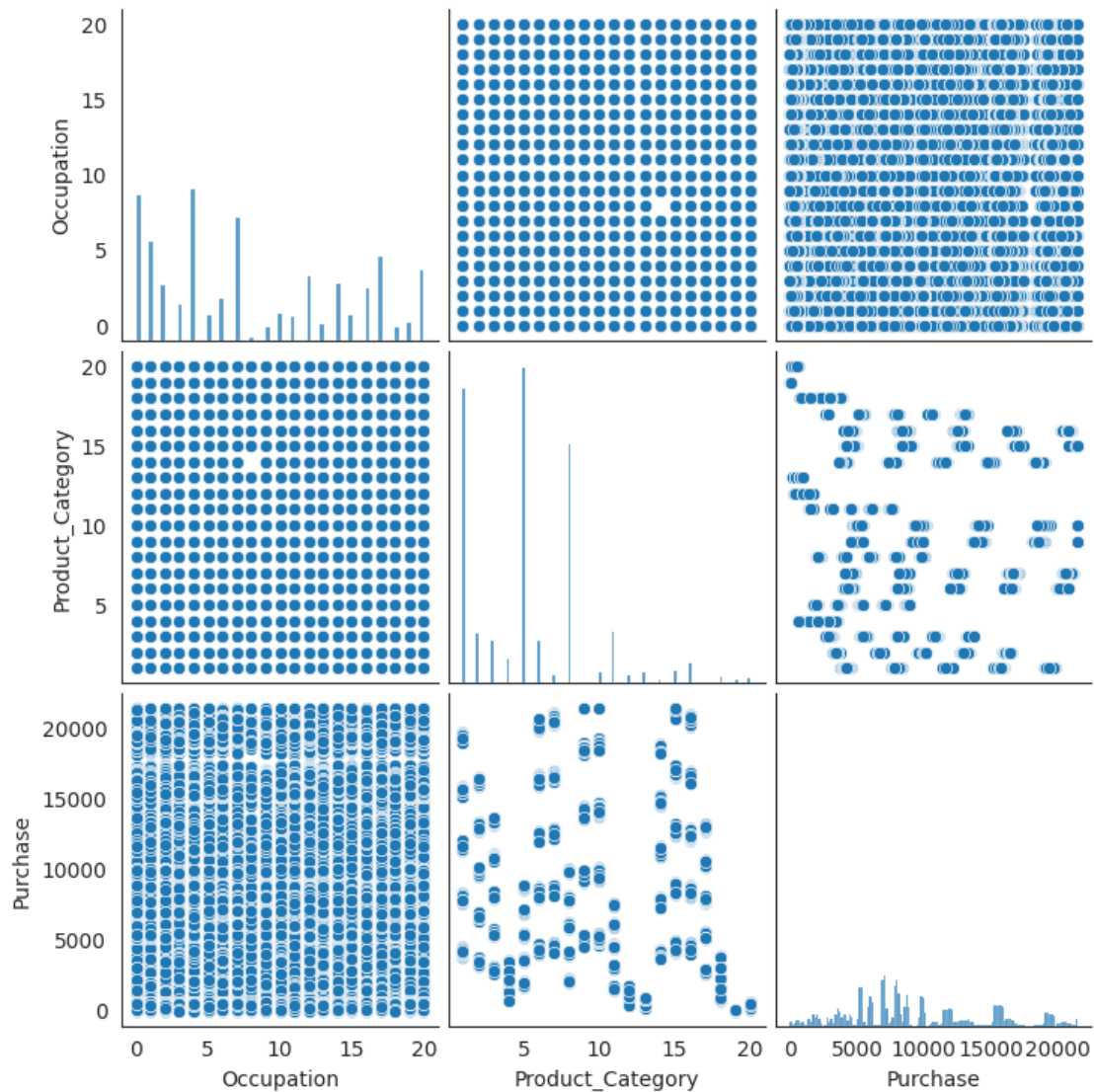


Correlation is a statistical measure that describes the strength and direction of a relationship between two variables.

```
plt.figure(figsize = (15, 5))
sns.heatmap(data=df.corr(), annot=True, cmap="cividis", square=True)
plt.show()
```



```
sns.pairplot(df_new)
plt.show()
```



Insights

- Our dataset is highly categorical variable centric. So we can hardly find correlation in our dataset.

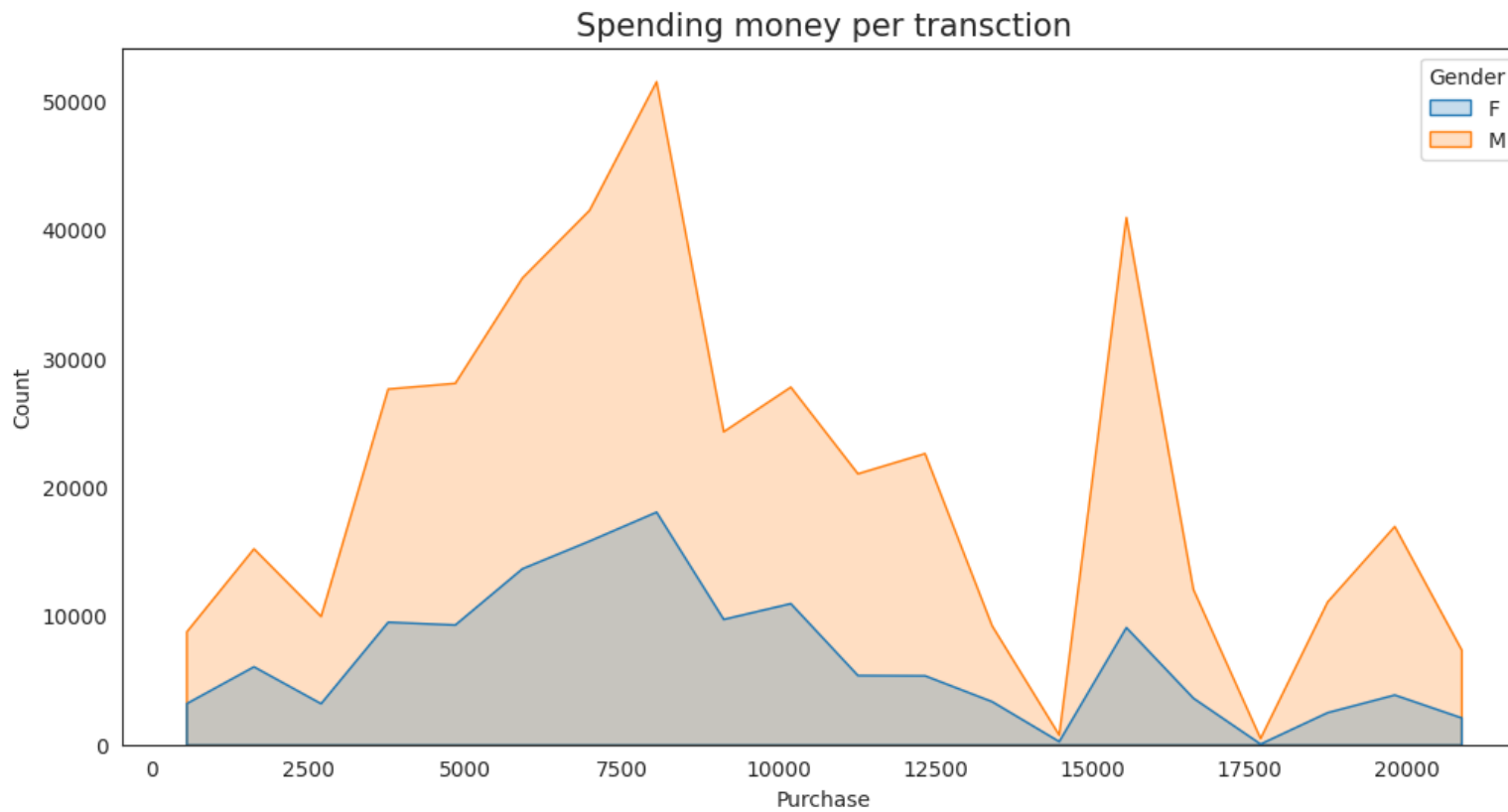
✧ 4. Answering questions

✧ 4.1 Are women spending more money per transaction than men? Why or Why not?

```
plt.figure(figsize=(12, 6))
```

```
plt.title("Spending money per transction", fontsize = 15)  
sns.histplot(data=df_new, x = "Purchase", bins=20, hue = "Gender", element="poly")
```

```
plt.show()
```



Insights

- The amount of money spent by women customers per transaction is quite less than that of men.

4.2 Confidence intervals and distribution of the mean of the expenses by female and male customers

```
df.head()
```

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase	
0	1000001	P00069042	F	0-17	10	A	2	0	3	8370	
1	1000001	P00248942	F	0-17	10	A	2	0	1	15200	
2	1000001	P00087842	F	0-17	10	A	2	0	12	1422	

```
amt_df = df.groupby(['User_ID', 'Gender'])[['Purchase']].sum()
amt_df = amt_df.reset_index()
amt_df
```

	User_ID	Gender	Purchase	
0	1000001	F	334093	
1	1000001	M	0	
2	1000002	F	0	
3	1000002	M	810472	
4	1000003	F	0	
...	...	...	...	
11777	1006038	M	0	
11778	1006039	F	590319	
11779	1006039	M	0	
11780	1006040	F	0	
11781	1006040	M	1653299	

11782 rows × 3 columns

Next steps:

Generate code with amt_df

☒ View recommended plots

```
male_df = amt_df[amt_df['Gender']=='M']
female_df = amt_df[amt_df['Gender']=='F']
amt_df['Gender'].value_counts()
```

```
F    5891
M    5891
Name: Gender, dtype: int64
```

```

genders = ["M", "F"]

male_sample_size = 3000
female_sample_size = 1500
num_repitions = 1000
male_means = []
female_means = []

for _ in range(num_repitions):
    male_mean = male_df.sample(male_sample_size, replace=True)['Purchase'].mean()
    female_mean = female_df.sample(female_sample_size, replace=True)['Purchase'].mean()

    male_means.append(male_mean)
    female_means.append(female_mean)

```

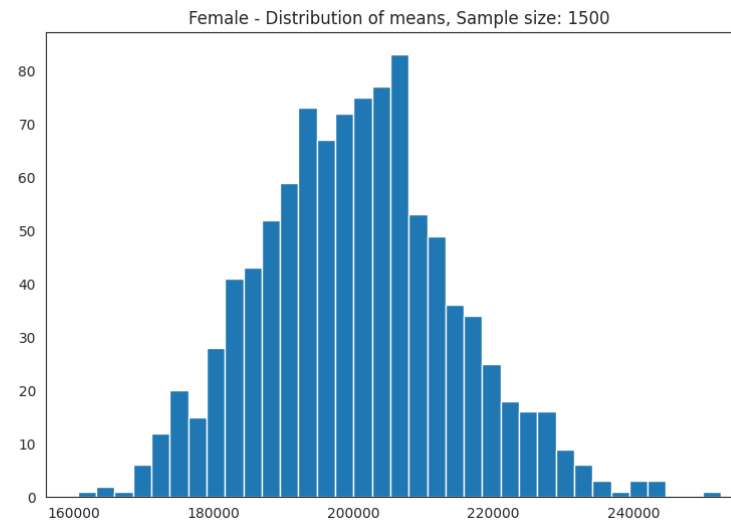
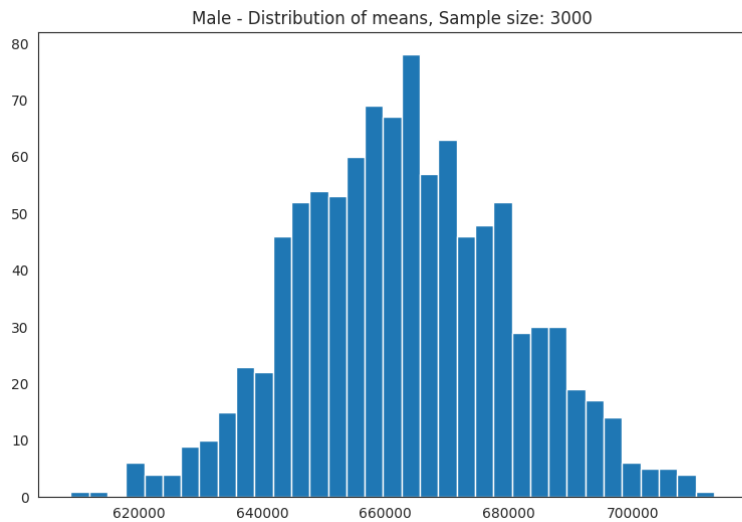
```

fig, axis = plt.subplots(nrows=1, ncols=2, figsize=(20, 6))

axis[0].hist(male_means, bins=35)
axis[1].hist(female_means, bins=35)
axis[0].set_title("Male - Distribution of means, Sample size: 3000")
axis[1].set_title("Female - Distribution of means, Sample size: 1500")

plt.show()

```



```

print("Population mean - Mean of sample means of amount spend for Male: {:.2f}".format(np.mean(male_means)))
print("Population mean - Mean of sample means of amount spend for Female: {:.2f}".format(np.mean(female_means)))

print("\nMale - Sample mean: {:.2f} Sample std: {:.2f}".format(male_df['Purchase'].mean(), male_df['Purchase'].std()))
print("Female - Sample mean: {:.2f} Sample std: {:.2f}".format(female_df['Purchase'].mean(), female_df['Purchase'].std()))

Population mean - Mean of sample means of amount spend for Male: 663178.93
Population mean - Mean of sample means of amount spend for Female: 200412.35

Male - Sample mean: 663653.05 Sample std: 933096.80
Female - Sample mean: 201363.54 Sample std: 535828.17

```

Observation

Now using the **Central Limit Theorem** for the **population** we can say that:

1. Average amount spend by male customers is **9,26,341.86**
2. Average amount spend by female customers is **7,11,704.09**

```

male_margin_of_error_clt = 1.96*male_df['Purchase'].std()/np.sqrt(len(male_df))
male_sample_mean = male_df['Purchase'].mean()
male_lower_lim = male_sample_mean - male_margin_of_error_clt
male_upper_lim = male_sample_mean + male_margin_of_error_clt

female_margin_of_error_clt = 1.96*female_df['Purchase'].std()/np.sqrt(len(female_df))
female_sample_mean = female_df['Purchase'].mean()
female_lower_lim = female_sample_mean - female_margin_of_error_clt
female_upper_lim = female_sample_mean + female_margin_of_error_clt

print("Male confidence interval of means: ({:.2f}, {:.2f})".format(male_lower_lim, male_upper_lim))
print("Female confidence interval of means: ({:.2f}, {:.2f})".format(female_lower_lim, female_upper_lim))

Male confidence interval of means: (639825.01, 687481.08)
Female confidence interval of means: (187680.36, 215046.73)

```

Now we can infer about the population that, **95% of the times**:

1. Average amount spend by male customer will lie in between: **(895617.83, 955070.97)**
2. Average amount spend by female customer will lie in between: **(673254.77, 750794.02)**

✓ Doing the same activity for married vs unmarried

```

amt_df = df.groupby(['User_ID', 'Marital_Status'])['Purchase'].sum()
amt_df = amt_df.reset_index()
amt_df.head()

```

	User_ID	Marital_Status	Purchase	
0	1000001	0	334093	
1	1000001	1	0	
2	1000002	0	810472	

Next steps:

[Generate code with amt_df](#)

☐

[View recommended plots](#)

```
amt_df['Marital_Status'].value_counts()
```

```
0    5891
1    5891
Name: Marital_Status, dtype: int64
```

```
marid_samp_size = 3000
unmarid_sample_size = 2000
num_repitions = 1000
marid_means = []
unmarid_means = []

for _ in range(num_repitions):
    marid_mean = amt_df[amt_df['Marital_Status']==1].sample(marid_samp_size, replace=True)['Purchase'].mean()
    unmarid_mean = amt_df[amt_df['Marital_Status']==0].sample(unmarid_sample_size, replace=True)['Purchase'].mean()

    marid_means.append(marid_mean)
    unmarid_means.append(unmarid_mean)
```

```
fig, axis = plt.subplots(nrows=1, ncols=2, figsize=(20, 6))
```